

Outcomes following Notched Ruthenium-106 Plaque Brachytherapy for Juxtapapillary Choroidal Melanomas

Manvi Manu Sobti^a Magdalena Edington^a Julie Connolly^a
David J. McLernon^b Stefano Schipani^a Diana Ritchie^a Paul Cauchi^a
Vikas Chadha^a

^aScottish Ocular Oncology Centre, Gartnavel General Hospital, Glasgow, UK; ^bUniversity of Aberdeen, Aberdeen, UK

Keywords

Juxtapapillary choroidal melanoma · COB plaque · Notched plaque · Ruthenium-106 · Brachytherapy

Abstract

Purpose: This study aimed to evaluate the outcomes of juxtapapillary choroidal melanomas treated with notched ruthenium-106 plaques. **Methods:** Juxtapapillary choroidal melanomas (tumours within 2 disc diameters from the optic disc) treated with notched ruthenium-106 plaques (Eckert & Ziegler, BEBIG, Berlin, Germany) at the Scottish Ocular Oncology Service between 2009 and 2015 were retrospectively reviewed. The data were analysed with respect to various outcome measures including recurrence, complications, vision, and eye preservation. **Results:** We reviewed 40 patients with a median tumour diameter of 8.4 mm (range 5–17 mm) and a median thickness of 2.5 mm (range 1.1–6 mm). AJCC tumour category distribution was 62.5% T1, 32.5% T2, and 5% T3 tumours. The mean presenting vision was 0.3 logMAR, and the mean final vision was 0.7 logMAR, with 62.5% retaining >1.0 logMAR and 50% retaining >0.3 logMAR at the final follow-up. The median follow-up was 51 months (14–100 months). Over the maximum follow-up time, 13 tumours

(32.5%) recurred. Six of these were treated with salvage proton beam therapy (PBT), 2 with transpupillary thermotherapy followed by PBT, and 5 with enucleation. The final eye retention rate was 87.5%. Complications included maculopathy (10%), retinal detachment (5%), neovascular glaucoma (2.5%), and diplopia (2.5%). The observed risk of recurrence over 5 years was 31% (95% CI: 14.1%, 47.8%), and the risk of enucleation over 5 years was 11.5% (95% CI: 0.9%, 21.8%). **Conclusion:** Juxtapapillary choroidal melanomas treated with notched ruthenium plaques have a high recurrence rate and frequently need salvage treatment with PBT for tumour control. This has led to a change in our practice toward offering PBT as the first-line treatment for these patients.

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Introduction

The treatment of juxtapapillary choroidal melanomas is challenging as any treatment involves radiation to visually crucial structures – the optic nerve and macula. These tumours are also known to have high recurrence rates (0–50%) [1–3]. Various eye-conserving radiation treatments like plaque brachytherapy, proton beam therapy

(PBT), transpupillary thermotherapy (TTT), and stereotactic radiosurgery have been used; however, there are no randomized control trials to determine the optimal form of treatment. Ruthenium-106 is a beta-emitting radioisotope widely used for delivering episcleral brachytherapy in Europe [4]. The plaque is placed over the tumour with a 2- to 3-mm margin to ensure the entire lesion receives the therapeutic radiation dose. Also, placement of the plaque needs to be precise as ruthenium has a very steep drop-off radiation dose [4, 5]. For tumours adjacent or near the optic disc, the nerve serves as an anatomical barrier for plaque placement, and thus plaques with a notch are used to accommodate the optic nerve and deliver radiation around it.

Plaque brachytherapy has the advantage of lower radiation collateral damage to the anterior segment compared to other conservative treatments like PBT [6, 7]. This is desirable as it means better vision preservation, less chance of a painful blind eye, and lower rate of enucleation for neovascular complications [8]. At the Scottish Ocular Oncology Service, our strategy for juxtapapillary choroidal melanomas had been to offer plaque brachytherapy for small- and medium-sized choroidal melanomas (<16-mm diameter and <6-mm height) as the first line of management and reserve TTT or PBT for salvage treatment for any recurrences. This retrospective review evaluated outcomes following notched ruthenium-106 plaque brachytherapy for juxtapapillary choroidal melanoma at our centre.

Methods

All juxtapapillary choroidal melanomas treated at the Scottish Ocular Oncology Service from January 2009 to December 2015 were retrospectively reviewed. Juxtapapillary tumours were defined as tumours touching the optic disc or with a posterior edge <2 disc diameters from the optic disc margin. Preoperative assessment included best-corrected visual acuity, clinical features (symptoms, laterality, location, distance from the optic disc, pigmentation, and presence of retinal detachment), fundus photographs, ultrasound scans (dimensions, shape, and features of the tumour), and metastasis screening (review of symptoms, chest X-ray, and liver ultrasound).

The surgical technique consisted of a conjunctival peritomy and localization of the tumour with an angled fibre-optic light probe and indirect ophthalmoscopy. A Ruthenium-106 COB plaque (19.8-mm diameter; Eckert & Ziegler, BEBIG, Berlin, Germany) was placed over the sclera, and the notched edge advanced posteriorly to accommodate the optic nerve. The plaque was sutured onto the sclera with 5-0 Dacron and the conjunctiva closed with 8-0 vicryl. The plaque was left in situ for a specified duration in order to deliver a scleral superficial prescribed dose of 470 Gy at the centre of the plaque. The dose fall at the notched edge was 24%

Table 1. Preoperative tumour characteristics

Lens status, %	
Phakic: 37	92.5
Pseudophakic: 3	7.5
Symptoms, %	
Asymptomatic: 20	50
Vision loss: 16	40
Metamorphopsia: 3	7.5
Pain: 1	2.5
Shape, %	
Dome: 32	80
Dome with tail: 4	10
Collar stud: 4	10
Pigmentation, %	
Pigmented: 35	87.5
Amelanotic: 5	12.5
Retinal detachment, %	
No: 35	87.5
Yes: 5	12.5
Distance from optic disc, %	
Touching optic disc margin: 24	60
Within 1 disc diameter from disc margin: 13	32.5
Within 2 disc diameters from disc margin: 3	7.5

(i.e., 112.8 Gy) of the dose prescribed at the centre of the COB plaque. Postoperatively, patients were reviewed at 6 weeks, 4 monthly for 1 year, and 6 monthly thereafter.

At each follow-up, visual acuity, clinical examination, and ultrasound scans were performed. Fundus photographs were taken to document any growth. Tumour control was indicated by regression or stabilization of tumour dimensions. Recurrence was defined by increase in tumour dimensions measured by ultrasound or by visible tumour growth along the disc margin on comparing serial photographs. Treatment for recurrence was decided as per tumour dimensions, visual potential, and patients' wishes to retain the eye. For patients requiring PBT, tantalum markers were inserted at our centre, and PBT was delivered at the Clatterbridge Cancer Centre. TTT or enucleation was carried out at our centre. Metastasis surveillance (review of symptoms and liver ultrasound) was carried out 6 monthly.

Outcome measures were tumour recurrence, complications, vision, and eye preservation. The chance of recurrence at 5 years was calculated using Kaplan-Meier analysis. The median (25th percentile, 75th percentile) follow-up time was estimated using the reverse Kaplan-Meier method.

The local ethics committee ruled that approval was not required for this study. The Tenets of the Declaration of Helsinki were followed while performing the study.

Results

During the study period, 40 patients with a median age of 64.5 (31–68) years were treated with ruthenium-106 plaque brachytherapy. This included 18 (45%) males and

Table 2. Management of tumour recurrence

	Type of recurrence	Time to recurrence, months	Treatment for recurrence
1	Increase in height	63	Enucleation
2	Increase in height	16	PBT
3	Increase in height	78	Enucleation
4	Increase in height	32	Enucleation
5	Increase in height	22	Enucleation
6	Increase in height	10	PBT
7	Increase in height	49	Two TTT treatments followed by further growth 2.5 years later that was treated with PBT. Three months following PBT, retrolental extension detected. Enucleation recommended but the procedure cancelled due to systemic problems
8	Increase in height	16	Enucleation
9	Growth at posterior edge	74	PBT
10	Growth at posterior edge	18	TTT and PBT 6 months later for further tumour growth
11	Growth at posterior edge	35	PBT
12	Growth at posterior edge	35	PBT
13	Growth at posterior edge	7	PBT

PBT, proton beam therapy; TTT, transpupillary thermotherapy.



Fig. 1. Colour fundus photograph demonstrates a juxtapapillary choroidal melanoma (a) and recurrence at the optic disc margin (b).

22 (55%) females. The right eye was involved in 52.5% ($n = 21$) and the left eye in 47.5% ($n = 19$) of the cases. Preoperative tumour characteristics are presented in Table 1. The median tumour diameter was 8.4 mm (range 5–17 mm), and the median thickness was 2.5 mm (range 1.1–6 mm). The AJCC (8th edition) tumour category distribution was 62.5% ($n = 25$) T1, 32.5% ($n = 13$) T2, and 5% ($n = 2$) T3 tumours. The mean presenting vision was

0.3 logMAR, and the mean final vision was 0.7 logMAR. The median (75th, 25th percentile) follow-up time was 51 months (28, 82 months). 22.5% of the patients lost >5 lines of vision, whereas 62.5% retained a vision >1.0 logMAR and 50% of the patients retained >0.3 logMAR.

Recurrence was seen in 32.5% (13 patients) over a maximum follow-up of 100 months. This included 8 patients (20%) with an increase in tumour height after ini-

tial shrinkage and 5 (12.5%) patients with an increase in basal diameter and growth along the optic disc margin (shown in Fig. 1). The median time to recurrence was 78 months (25th percentile, 35 months; 75th percentile undefined as not enough events). The management of recurrences is summarized in Table 2. Two recurrences (5%) were treated with TTT, but unfortunately these tumours continued to grow and required additional PBT. PBT achieved tumour control in 1 patient, but the other patient developed a retroscleral extension. Although enucleation was recommended for this patient, it could not go ahead due to comorbidities. PBT alone was used in 6 patients (15%) and enucleation alone in 5 patients (12.5%). Complications included maculopathy in 4 patients (10%), retinal detachment in 2 (5%), neovascular glaucoma in 1 (2.5%), and diplopia in 1 (2.5%).

The eye retention rate was 87.5%. Metastases were recorded in 7 (17.5%) patients. Of these, 4 patients had liver metastasis, 1 had lung metastasis, and 1 had multiple (liver, lung, and bone) metastasis. Out of the 7 patients who developed metastasis, 3 patients had suffered a local recurrence of the choroidal melanoma. The time interval between the treatment and metastasis was 16–115 months (median 47 months). All-cause mortality was 17.5% over the maximum follow-up time of 100 months. The risk of local recurrence within 5 years was 31% (95% CI: 14.1%, 47.8%) (shown in Fig. 2). The risk of enucleation within 5 years was 11.5% (95% CI: 0.9%, 21.8%). The risk of all-cause mortality within 5 years was 25.1% (95% CI: 6.7%, 43.5%).

Discussion

A recurrence rate of 31% at 5 years for juxtapapillary choroidal melanomas in our series was high, whereas outcomes for complications, vision, and eye preservation were favourable. Recurrence rates of 0–50% for juxtapapillary choroidal melanomas [3] have been reported, and although comparison is difficult due to the use of various radioisotopes and adjuvant TTT, our series has a high recurrence rate.

Our strategy of offering plaque brachytherapy as the first-line treatment for small- to medium-sized juxtapapillary choroidal melanomas was based on good tumour control (>80% control) seen in posterior pole melanomas and low incidence of complications with ruthenium plaques [2, 8]. Other modalities like TTT and PBT were reserved to treat tumour recurrences. This conservative approach was adopted to preserve vision and avoid enu-

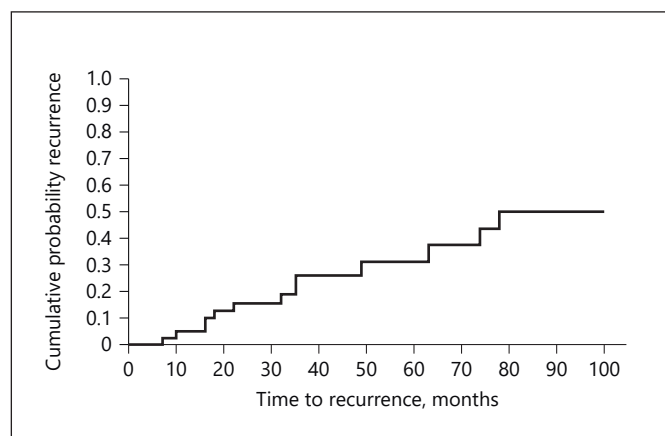


Fig. 2. Kaplan-Meier plot of the cumulative probability of recurrence over time in months.

cleation for painful blind eyes. The COMS trial demonstrated no difference in survival rates between radiation and enucleation for medium-sized choroidal melanomas [9]; however, the trial did not analyse radiotherapy for juxtapapillary tumours. Equivalent survival between plaque radiotherapy and enucleation for juxtapapillary tumours has been reported [10]. Plaque brachytherapy is thus a viable option, especially for small- to medium-sized tumours.

Tumour control of 90% with ruthenium plaques has been reported; however, 30% of the patients had received additional xenon-arc laser [1]. Local control reported with iodine and palladium plaques has been much higher compared to our series. In a large series of 650 juxtapapillary choroidal melanomas treated with various plaques (iodine, palladium, ruthenium, and iridium) and adjuvant TTT, recurrence rates of 14% and 21% at 5 and 10 years, respectively, were reported [7]. In addition, plaques with a slot for accommodating the optic nerve have been reported to have 98.1% tumour control [11]. We have not seen similar results with notched ruthenium plaques. There may be several reasons contributing to this high recurrence.

First, the optic nerve anatomy poses a unique challenge even with the use of notched plaques. The optic nerve head diameter measures about 1.8 mm on average. As the nerve leaves the globe, it is enveloped by the myelin sheath, and its diameter at this section is 5–6 mm [11]. The notched plaque surrounds this myelinated section of the nerve, and thus even with accurate placement of the plaque abutting the nerve, there may be a 2-mm region adjacent to the optic nerve head that does not lie in the

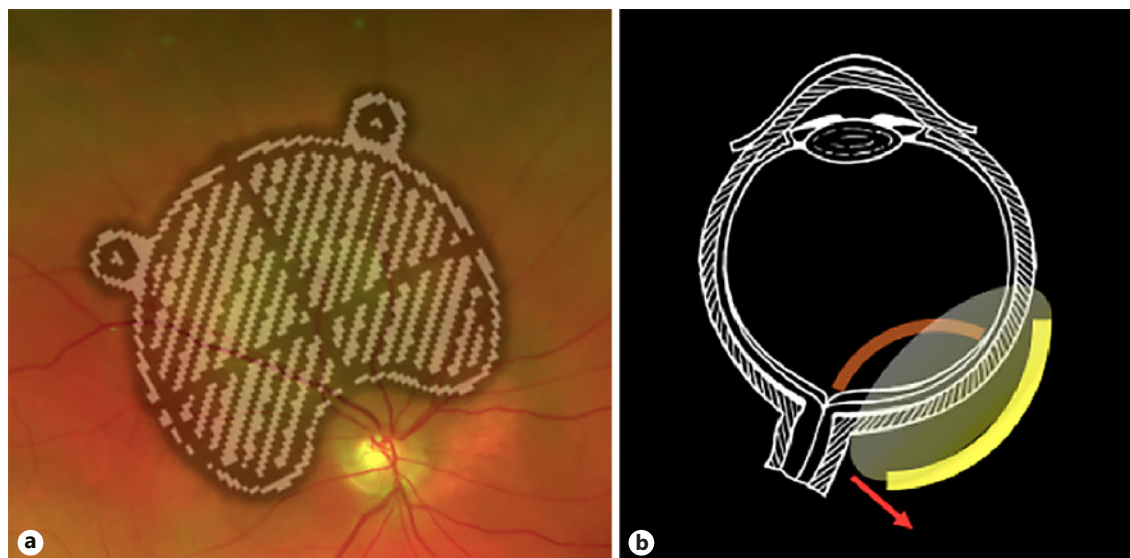


Fig. 3. Diagrammatic representation of the position of the notched ruthenium-106 plaque in relation to the optic nerve (a) and tilt of the posterior edge of the plaque (b).

radiation field (shown in Fig. 3). In addition to this, the COB plaque has a 0.7-mm inactive edge [12]. This side wall further shields the tumour from the radiation [13]. The notched portion of the COB plaque has a dose fall to 24% according to the manufacturer's specifications. All these factors contribute to the risk of irradiation geographical miss for tumours in close proximity to the optic nerve.

Second, notched plaques are surgically difficult to insert. The posterior edge of the plaque cannot be placed under direct vision, and dissection in the posterior orbit is challenging. There can be an "edge lift off" or tilt, and this may reduce the effective dose of radiation to the tumour [8] (shown in Fig. 3). The impact of this factor may be reduced by the use of intraoperative ultrasound (which is not routinely used in our service).

Third, the radiation field of ruthenium plaques differs from other radioisotopes. Ruthenium plaques have a very sharp penumbra due to the rapid drop-off of radiation dose [14]. If the plaque is not accurately placed, there is a high risk of a geographical miss. In addition, the radiation field does not extend to fill-in the volume within the notch [13]. For iodine and palladium plaques, the radioactive component is in the form of small radioactive seeds. The strength and position of these seeds can be customized to conform to the location and size of the tumour [15]. This ensures the entire tumour receives the therapeutic dose. Finger's slotted eye plaques with an 8-mm wide slot have been used. These not only accom-

modate the entire optic nerve sheath but also extend beyond the nerve to the opposite side of the disc and allow the entire tumour and a safety zone to be included in the radiation field. Further, by customizing the strength and position of the radioactive seeds, it is possible to "fill-in" the radiation dose within the slot. Ruthenium plaques on the other hand are solid source plaques and cannot be customized or cut. Dose distribution simulations have shown that compared with circular plaques, the COB plaques have an asymmetrical dose distribution and also have a smaller zone of therapeutic dose [4].

Of the 13 recurrences, 5 patients had additional tumour growth at the optic disc margin. This represents an obvious geographical miss in the area adjacent to the optic nerve and highlights the difficulty in treating these tumours.

The use of TTT along with brachytherapy to treat juxtapapillary choroidal melanomas is well established [16]. However, the 2 recurrences treated with TTT in our series had further tumour growth and eventually required PBT. One patient developed a retroscleral extension 3 years after the TTT. This complication has been reported in the literature [17, 18]. This highlights the risk of TTT and the need to follow these patients very closely. PBT alone was used in 6 patients (15%) and enucleation alone in 5 patients (12.5%). Thus, majority of our recurrences received PBT as the secondary treatment.

The rate of complications in our series was low. Neovascular glaucoma (NVG) was seen in only 1 case (2.5%)

and was managed conservatively. This contrasts with higher complications reported with other plaques with NVG seen in 11% with slotted palladium plaques at 48 months, 36% with helium ion radiation at 42 months [17], and 33% with carbon ion radiation at 55 months [18]. An incidence of NVG of 15% at 5 years and 22% at 10 years was reported with the use of various plaques [7].

All the enucleations in our series were for tumour recurrence rather than to manage painful blind eyes. Thus, eye retention was high in our series compared to that reported in the literature. Secondary enucleation rates have ranged from 6 to 40% [19]. An enucleation rate of 23% has been reported, with 52% performed for NVG, 11% for ocular pain, 11% for vitreous haemorrhage, and 26% for tumour recurrence [19].

Visual outcomes were favourable. Only 37% patients had worse than 1.0 logMAR vision at final follow-up and 22.5% lost >5 lines of vision. This is in contrast to a series of juxtapapillary choroidal melanomas overhanging the disc where 77% patients had worse than 6/60 vision and 63% had lost >5 lines of vision. In a large series of 650 patients, a vision of worse than 6/60 was seen in 54% cases at 5 years and in 87% at 10 years and loss of >5 lines of vision was seen in 45% and in 78% at 5 and 10 years, respectively [7].

Ruthenium-106 has been reported to have lower tumour control compared to other plaques but also lower incidence of radiation-related side effects [1]. We have seen a similar trend in our series. Local treatment failure after brachytherapy in the COMS trial had demonstrated a trend toward increased metastasis risk and reduced survival [20, 21]. This prompted a change in our practice to improve the tumour control. There are no randomized controlled trials comparing PBT and plaque brachytherapy; however, various retrospective series have shown better tumour control with PBT [22]. In a review of all globe-conserving treatments for choroidal melanomas, the local failure rate of PBT was 4.2% as opposed to 9.6% average for brachytherapy from all radioisotopes [23]. In our previously published series on PBT for choroidal and ciliary body melanomas, patients achieved a 5-year disease-specific survival rate of 87.7% and an actuarial 5-year eye retention rate of 71.3% [24]. In a study comparing stereotactic radiosurgery and PBT for choroidal melanoma over a 10-year period, both modalities had comparable tumour control and eye retention. However, patients treated with PBT retained better vision postoperatively [25]. We have thus changed our practice to offering PBT as the first-line treatment for small- to medium-sized juxtapapillary tumours. Patients are counselled regarding the higher incidence of radiation-related side effects to the anterior seg-

ment [6] such as dry eye, loss of lashes, radiation cataract, NVG, and risk of secondary enucleation.

Limitations of this study include its retrospective design and small population size. However, as all choroidal melanomas in Scotland are treated at our centre, the number of cases represents the incidence of this rare tumour.

Conclusion

The recurrence rate of juxtapapillary choroidal melanomas treated with notched ruthenium plaques is high. These tumours frequently require salvage treatment for tumour control. Vision and eye preservation on the other hand are favourable. We, therefore, now offer PBT as the first-line treatment for small- to medium-sized choroidal melanomas within 2 disc diameters from the optic disc margin.

Acknowledgment

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Statement of Ethics

As this was a retrospective study to evaluate results of our standard practice, the West of Scotland Research Ethics Service confirmed that ethical approval was not required. The Tenets of the Declaration of Helsinki were followed while performing the study. Written informed consent was obtained for all patients for the treatment and use of anonymized medical images for teaching and publication. The study had been approved by the Caldicott Guardian at NHS Greater Glasgow and Clyde.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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None.

Author Contributions

P.C., S.S., and V.C. made substantial contributions to conception and design of work. M.E., J.L., M.S., D.M., P.C., and V.C. contributed to data acquisition, analysis, and interpretation of data.

M.S., M.E., J.L., S.S., D.M., D.R., P.C., and V.C. contributed to drafting of the work or revising it critically for important intellectual content. M.S., M.E., P.C., and V.C. gave final approval of the version published. Illustrations were done by M.S. All agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Data Availability Statement

The data that support the findings of this study are not publicly available as the data belong to NHS Greater Clyde but are available from the corresponding author (M.S.) upon reasonable request. Permission from the Caldicott Guardian was obtained for the purpose of the study.

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